

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.5: Area of Polygons — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.5: Area of Polygons
Driving Question	DRIVING QUESTION: How can we calculate the area of any polygon — no matter how many sides or how irregular — so that we can fairly divide the Kericho community land plot and solve other real Kenyan land and design problems?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring the Phenomenon — The Kericho Land Plot: How Much Land Is There, and Can We Share It Fairly?	Students examined a scaled diagram of the Kericho community land plot — an irregular triangular section — and estimated its area using a square-centimetre grid overlay. They observed that counting whole and partial grid squares gave only a rough area, and that slanted sides made exact counting unreliable. Key pedagogical note: the tension between the grid estimate and the need for precision is the productive frustration that motivates the formula. Teachers should record all student estimates on the board and highlight the wide range of answers as evidence that a better method is needed.	Students learned that the area of a triangle is calculated using $A = \frac{1}{2}bh$, where b is the length of the chosen base and h is the PERPENDICULAR height — the shortest distance from the base line to the opposite vertex, measured at a right angle. They learned that h is NOT the slant (oblique) side of the triangle. Teacher note: emphasise with a physical demonstration — hold a ruler flat as the base, then drop a set-square vertically to show the perpendicular height. Students also learned how to identify the correct base–height pair when the triangle is acute, right-angled, or obtuse (where h falls outside the triangle).	The formula $A = \frac{1}{2}bh$ is explained by the fact that any triangle is exactly half of a parallelogram (or rectangle) with the same base and perpendicular height. Teachers should show learners how to duplicate the triangle, rotate the copy 180° , and join the two to form a parallelogram, making the halving visually obvious. In the Kericho context: if the triangular plot has base 40 m and perpendicular height 25 m, then $A = \frac{1}{2} \times 40 \times 25 = 500 \text{ m}^2$ — a precise answer replacing the unreliable grid estimate. This result anchors the driving question: if we can find the area of one triangle, can we use triangles to find the area of any polygon?
Lesson 2 Quadrilaterals — Parallelograms, Rhombuses, Trapeziums and Kites	Students were given cut-out shapes of a parallelogram, a rhombus, a trapezium, and a kite (each labelled with Kenyan land/design contexts — a maize-farm parallelogram plot in Rift Valley, a rhombus tile at a Nairobi hotel, a trapezium-shaped shamba in Kisumu, and a kite-shaped garden in Mombasa). They measured diagonals and perpendicular heights using	Students derived and recorded four area formulae: (1) Parallelogram: $A = bh$, where h is the perpendicular height between the parallel sides — not the slant side. (2) Rhombus: $A = \frac{1}{2}d_1d_2$, where d_1 and d_2 are the two diagonals that bisect each other at right angles. (3) Trapezium: $A = \frac{1}{2}(a + b)h$, where a and b are the two parallel sides and h is the perpendicular distance	Each formula is explained by decomposition into triangles or rectangles: the parallelogram rearranges into a rectangle ($A = bh$); the rhombus and kite split into four right triangles whose combined area gives $\frac{1}{2}d_1d_2$; the trapezium is the sum of two triangles sharing the same height, with bases a and b , giving $\frac{1}{2}ah + \frac{1}{2}bh = \frac{1}{2}(a+b)h$. Teachers should connect each

	rulers and observed how each shape could be physically rearranged or cut into rectangles and triangles. Teacher note: direct students to cut the parallelogram along a perpendicular height and slide the triangle to form a rectangle — this observable transformation is the conceptual core of the lesson.	between them. (4) Kite: $A = \frac{1}{2}d_1d_2$, where d_1 and d_2 are the diagonals (one of which is the axis of symmetry). Teacher note: students often confuse the slant height with the perpendicular height for the parallelogram and trapezium — use a red pen to mark h explicitly on all diagrams.	formula back to the Kericho plot: if a community decides to subdivide the land into a trapezoidal plot for a family and a parallelogram plot for a cooperative, both areas can now be calculated precisely. Pedagogical note: ask students, 'Which formula would you use if you only had a measuring tape and could only measure the diagonals?' — this highlights the practical value of the rhombus/kite formula for irregular land measurement.
Lesson 3 Pentagons and Hexagons — Decomposing the Kericho Plot's Neighbouring Parcels	Students examined scaled diagrams of a pentagonal plot (representing a five-sided community garden near Kericho) and a hexagonal plot (representing a six-sided cooperative farm). They were challenged to find the area without a new formula. Using rulers and pencils, students drew diagonals from one chosen vertex to all non-adjacent vertices, observing that the pentagon split into exactly 3 triangles and the hexagon into exactly 4 triangles. Teacher note: circulate and ensure students draw diagonals from a SINGLE vertex — this is the key constraint of the triangulation method. Record on the board: pentagon (5 sides) → 3 triangles; hexagon (6 sides) → 4 triangles. Ask: 'Do you see a pattern between the number of sides and the number of triangles?'	Students learned that any convex polygon with n sides can be decomposed into $(n - 2)$ non-overlapping triangles by drawing diagonals from one interior vertex to all other non-adjacent vertices. For a pentagon: $5 - 2 = 3$ triangles; for a hexagon: $6 - 2 = 4$ triangles. The total area is the sum of all individual triangle areas: $A = \frac{1}{2}b_1h_1 + \frac{1}{2}b_2h_2 + \dots + \frac{1}{2}b_{(n-2)}h_{(n-2)}$. Students also learned the formula for a REGULAR polygon: $A = \frac{1}{2} \times \text{perimeter} \times \text{apothem}$, where the apothem is the perpendicular distance from the centre to the midpoint of any side. Teacher note: for the regular hexagon specifically, $A = (3\sqrt{3}/2)s^2$, where s is the side length — worth noting as a shortcut for regular hexagonal plots.	The $(n - 2)$ triangle rule is explained by the fact that each diagonal drawn from one vertex 'uses up' one triangle without creating a new side count, and the two boundary sides of the polygon form the final triangle. Teachers should draw the general case on the board, labelling vertices V_1 through V_n and showing diagonals from V_1. In the Kericho context: a pentagonal community hall floor with sides and measured heights gives a precise area by summing three triangles — no new formula needed beyond $A = \frac{1}{2}bh$. Pedagogical note: explicitly connect this lesson to Lesson 1 — 'We are still using the triangle area formula from Day 1; we are just applying it more times.' This reinforces the cumulative model and prepares students for heptagons, octagons, and irregular polygons in later lessons.
Lesson 4 Heptagons — Decomposing the Kericho Land Plot into Triangles	Students were given a scaled diagram of a heptagonal (seven-sided) land plot representing an extended section of the Kericho community land. They drew diagonals from one chosen vertex and observed the plot splitting into exactly five non-overlapping triangles ($7 - 2 = 5$). They measured the base and perpendicular height of each triangle using a ruler and protractor. Teacher note: some students will choose different starting vertices — show that the total area remains the same regardless of which vertex is chosen, reinforcing that decomposition is a strategy, not a rigid procedure. Record all five base—	Students learned that the area of a heptagon equals the sum of the areas of five non-overlapping triangles formed by drawing diagonals from one interior vertex: $A = \frac{1}{2}b_1h_1 + \frac{1}{2}b_2h_2 + \frac{1}{2}b_3h_3 + \frac{1}{2}b_4h_4 + \frac{1}{2}b_5h_5$. They practised measuring perpendicular heights carefully, distinguishing them from slant sides. They also confirmed the general rule from Lesson 3: an n-sided polygon always yields $(n - 2)$ triangles. Teacher note: a common error is measuring the slant diagonal instead of the perpendicular height of each triangle — colour-code each triangle and its height in a different colour to reduce confusion.	The heptagon area calculation is explained as a direct extension of the triangulation principle established in Lessons 1 and 3. There is no new formula — only a repeated application of $A = \frac{1}{2}bh$ across five triangles. Teachers should reinforce this explicitly: 'We have not learned a new formula today; we have learned that one powerful formula ($A = \frac{1}{2}bh$) can unlock the area of any polygon when combined with decomposition.' In the Kericho driving question context: if the full community land plot is heptagonal, students can now calculate its total area and begin thinking about how to subdivide it fairly —

	height pairs on the board before any calculation begins.		connecting directly to the unit's central problem. Pedagogical note: ask, 'If you added an eighth side to this plot, how many triangles would you need?' to pre-load the octagon lesson.
Lesson 5 Octagons — Decomposing the Mombasa Courtyard	Students examined a scaled plan of a regular octagonal courtyard design from a Mombasa coastal hotel. They first decomposed it into six triangles using the diagonal-from-one-vertex method ($8 - 2 = 6$), measuring each triangle's base and height. They then observed a second method: the teacher demonstrated dividing the octagon into 8 congruent isosceles triangles from the centre point, each with base equal to one side length (s) and height equal to the apothem (a). Students compared the two methods and verified they gave identical total areas. Teacher note: the dual-method comparison is the pedagogical centrepiece — it shows students that mathematical flexibility (choosing the most efficient method) is a problem-solving skill.	Students learned two equivalent methods for finding the area of a regular octagon: (a) Triangulation from one vertex: decompose into 6 triangles, sum their areas. (b) Centre-triangle method: decompose into 8 congruent isosceles triangles, $A = 8 \times (\frac{1}{2} \times s \times a)$, which simplifies to $A = \frac{1}{2} \times \text{perimeter} \times \text{apothem} = \frac{1}{2}Pa$. They also learned the direct formula for a regular octagon with side length s : $A = 2(1 + \sqrt{2})s^2$. Teacher note: the formula $A = \frac{1}{2}Pa$ (half the perimeter times the apothem) works for ANY regular polygon — flag this as a generalisation students should memorise and connect back to the regular hexagon from Lesson 3.	The formula $A = \frac{1}{2}Pa$ is explained by the fact that each of the n central triangles has area $\frac{1}{2} \times s \times a$, and summing n such triangles gives $n \times \frac{1}{2}sa = \frac{1}{2} \times (ns) \times a = \frac{1}{2}Pa$, since $ns = \text{perimeter}$. Teachers should write this derivation step by step on the board. In the Mombasa context: if the courtyard has side length 3 m, its perimeter is 24 m; if the apothem is approximately 3.62 m, then $A = \frac{1}{2} \times 24 \times 3.62 \approx 43.4 \text{ m}^2$. Connect to Kericho: 'If the community wanted to build an octagonal meeting hall on part of the Kericho plot, we can now calculate exactly how much land it would occupy.' Pedagogical note: ensure students understand that the apothem is always perpendicular to the side — this is the same perpendicular-height principle from Lesson 1 appearing in a new form.
Lesson 6 Irregular Polygons — Triangulation of the Nakuru School Shamba	Students were given a scaled map of an irregularly shaped school shamba (farm plot) at a secondary school in Nakuru. The plot had eight sides of different lengths and no right angles. Students identified one reference vertex and drew diagonals to all non-adjacent vertices, observing that the shamba decomposed into exactly six ($8 - 2$) non-overlapping triangles. They measured each triangle's base and used a set-square to construct and measure the perpendicular height. Teacher note: this is the first time students work with a fully irregular polygon where no sides are equal and no angles are standard. Expect difficulty in measuring perpendicular heights accurately — remind students to extend the base line if the foot of the perpendicular falls outside the triangle (obtuse triangles).	Students learned that any irregular polygon with n sides can be divided into exactly $(n - 2)$ non-overlapping triangles by drawing diagonals from one reference vertex to all non-adjacent vertices, and that this method works regardless of how irregular the shape is. Total area = sum of all $(n - 2)$ triangle areas. They also learned a practical surveying strategy used in Kenyan land measurement: labelling vertices systematically ($A, B, C, \dots$), recording all base and height measurements in a table before calculating, and summing triangle areas carefully to avoid omissions. Teacher note: the table method is a key organisational skill — model it explicitly on the board and require students to use it in their written work.	The universality of the triangulation method for irregular polygons is explained by the mathematical fact that any polygon (convex or concave, regular or irregular) can be triangulated, and the triangle is the simplest polygon whose area formula is exact and well-defined. Teachers should contrast this with the grid-counting method from Lesson 1: 'Grid counting was approximate and depended on the grid size; triangulation gives us an exact answer using only a ruler and the formula $A = \frac{1}{2}bh$.' In the Nakuru context: accurate land area is legally important for school land registration with county governments — making the mathematics directly relevant to Kenyan civic life. Connect to driving question: 'We now have the tool to handle the Kericho plot no matter what shape it is. What do we do if the plot is made up of several different shapes joined together?' — pre-loading Lesson 7.

Lesson 7 Composite Polygons and Real-Life Application: Solving the Kericho Land-Division Problem	Students received a composite diagram of the full Kericho community land plot, which combined a trapezoidal section, a rectangular section, and a triangular section joined together. They also examined a floor-plan design problem: a Nairobi architect's plan for a community centre whose floor combined a regular hexagonal hall and a rectangular entrance lobby. Students identified component shapes within each composite figure and observed that the shapes shared boundary lines but did not overlap. Teacher note: some students will attempt to use a single formula for the whole composite — guide them with the question, 'Is there a single polygon formula that covers this entire shape?' to surface the need for decomposition.	Students learned that the area of any composite polygon equals the sum of the areas of its non-overlapping component shapes (addition method), or, where a shape is a larger polygon with a piece removed, equals the area of the larger shape minus the area of the removed piece (subtraction method). They applied both methods: addition for the Kericho land plot (trapezium + rectangle + triangle) and subtraction for a design problem (rectangle minus a triangular cut-out). They also practised the full Kericho land-division problem: calculating the total area, then determining how to divide it into equal or proportional sub-plots for different community groups. Teacher note: connect to proportional reasoning — if one family is entitled to 40% of the plot, how many square metres is that?	The composite-area principle is explained by the additive property of area: if two regions do not overlap, the area of their union equals the sum of their individual areas. This is a foundational geometric property, not a new formula. Teachers should draw a Venn-style area diagram to show non-overlapping regions summing to a total. In the Kericho driving question context: this lesson is the direct solution lesson — students calculate the exact total area of the community land plot and produce a fair-division plan with justified area calculations for each sub-plot. Pedagogical note: require students to present their division plan as a scaled diagram with area labels, simulating the kind of document a Kenyan land surveyor would produce for the county land registry.
Lesson 8 Model Building, DQB Resolution & Final Assessment — Solving the Kericho Land Division Problem	Students reviewed all entries on the Driving Question Board accumulated across Lessons 1–7 and identified which questions had been answered, which had evolved, and which remained open. They displayed their cumulative polygon area models — revised diagrams showing the progression from triangle → quadrilaterals → pentagons/hexagons → heptagons → octagons → irregular polygons → composite polygons. They also observed peer groups' final Kericho land-division proposals and compared different decomposition strategies that yielded the same total area. Teacher note: the DQB review is a metacognitive milestone — explicitly ask students, 'How has your answer to the driving question changed from Lesson 1 to today?' Record representative responses on the board.	Students synthesised the unit's central learning: any polygon — no matter how many sides or how irregular — can be decomposed into triangles and quadrilaterals whose areas can be calculated using $A = \frac{1}{2}bh$ (triangle), $A = bh$ (parallelogram), $A = \frac{1}{2}(a+b)h$ (trapezium), $A = \frac{1}{2}d_1d_2$ (rhombus/kite), and $A = \frac{1}{2}Pa$ (regular polygon). The total area of any polygon, including composite and irregular ones, is the sum of its component areas. Students also learned to present a complete, justified land-division solution: scaled diagram, area calculations, fair-division rationale, and a summary statement answering the driving question. Teacher note: the summary statement is an assessment of conceptual understanding — look for students who can articulate WHY triangulation works, not just HOW to perform it.	The full explanatory model is: (1) Identify the polygon type; (2) If standard, apply the appropriate formula directly; (3) If irregular or composite, decompose into triangles (and standard quadrilaterals where efficient); (4) Calculate each component area using $A = \frac{1}{2}bh$ or the appropriate quadrilateral formula; (5) Sum all component areas for the total. This model answers the driving question completely: we can calculate the area of ANY polygon by reducing it to triangles — the most fundamental polygon — and applying one core formula. Teachers should close the unit by returning to the very first Kericho diagram from Lesson 1 and showing how the class's answer has become precise, justified, and generalisable. Pedagogical note: celebrate the progression from 'rough grid estimate' to 'exact, professionally presented land-division plan' as evidence of genuine mathematical growth across the unit.